THRESHOLD ENTERPRISE SYSTEMS
AI Model Approval Policy

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1. CONFIDENTIAL CLASSIFICATION AND OPERATIONAL PURPOSE

*** CONFIDENTIAL DOCUMENT - RESTRICTED TO MANAGERIAL AND TECHNICAL GOVERNANCE ROLES ***

This document contains proprietary evaluation workflows, gateway approval criteria, technical documentation standards, and model certification protocols for Threshold Enterprise Systems. It is classified as CONFIDENTIAL. Access is restricted to authorized roles (MANAGER, AI_ENGINEER, SECURITY_ENGINEER, ADMIN). Workforce members holding the general EMPLOYEE role are not authorized to retrieve this document within the knowledge base.

The purpose of this AI Model Approval Policy is to establish a rigorous, mandatory, gate-controlled review and certification lifecycle that every artificial intelligence model, retrieval-augmented generation (RAG) system, fine-tuned large language model (LLM), and third-party AI integration must successfully navigate prior to production deployment.

By implementing standardized gating mechanisms, Threshold Enterprise Systems ensures that no algorithmic system is introduced into staging or production environments without exhaustive validation of its technical stability, safety, bias profile, privacy protections, and alignment with corporate strategic objectives.

2. Scope and Model Archetypes Covered

This policy applies to all AI artifacts developed internally, adapted, fine-tuned, or procured from external commercial vendors across all operating entities of Threshold Enterprise Systems in India, the United States, the United Kingdom, and Singapore.

The approval requirements govern five distinct model archetypes:

1. Custom Internally Trained Models: Foundation models, predictive machine learning models, and deep neural networks trained on corporate or customer datasets.

2. Fine-Tuned Large Language Models: Pre-trained foundation models adapted using parameter-efficient fine-tuning (PEFT, LoRA) or full weight fine-tuning on internal corpora.

3. Retrieval-Augmented Generation (RAG) Systems: Complex architectures coupling vector databases, embedding pipelines, retrieval filters, and generative models.

4. Autonomous AI Agents: Multi-step agentic systems capable of planning, tool invocation, code generation, or API execution.

5. Third-Party Commercial AI Services: External SaaS APIs (e.g., Anthropic, OpenAI, Google Cloud Vertex) integrated into enterprise applications.

3. The End-to-End Approval Lifecycle Workflow

Every AI initiative must progress sequentially through seven mandatory governance gates:

Stage 1 - AI Use Case Registration: The project sponsor registers the initiative in the Enterprise AI Registry, detailing business objectives, intended user base, and architectural overview.

Stage 2 - Risk Classification (AIG-002 Gate): The initiative is evaluated under the AI Risk Classification Framework (AIG-002), formally certifying its risk tier (LOW, MEDIUM, HIGH, or CRITICAL).

Stage 3 - Technical Review: The AI Engineering Architecture Review Board audits model performance benchmarks, evaluation datasets, inference latency, hardware resource profiles, and explainability mechanisms.

Stage 4 - Privacy Review (AIG-006 Gate): The Data Privacy Office verifies training data lineage, PII scrubbing, data minimization, and cross-border transfer compliance.

Stage 5 - Security Review (AIG-007 Gate): Information Security conducts adversarial red-team penetration testing, prompt injection resistance evaluations, and dependency vulnerability scans.

Stage 6 - Governance Approval: The Enterprise AI Ethics and Governance Board (AIEGB) reviews the assembled dossier, verifying alignment with Responsible AI principles (AIG-001).

Stage 7 - Deployment Authorization (AIG-010 Gate): Final formal sign-off granting promotion to live production staging.

4. Mandatory Documentation Dossier Requirements

Approval cannot be granted without the submission of a complete, standardized Governance Dossier cataloged in the registry:

- Standardized Model Card: Detailing model architecture, parameter scale, intended use cases, known limitations, and training hardware metrics.

- Data Sheet for Datasets: Documenting training and fine-tuning data provenance, data cleansing procedures, demographic distributions, and intellectual property licensing terms.

- Algorithmic Bias and Fairness Audit Report: Documenting demographic parity, equalized odds, and disparate impact ratios across protected personal classes.

- Adversarial Robustness and Prompt Security Assessment: Documenting results of prompt injection stress tests (AIG-007), jailbreak resilience, and context boundary verification.

- Human-in-the-Loop Operational Plan: Detailing reviewer qualifications, override workflows, and logging systems in compliance with AIG-005.

- Fallback and Decommissioning Runbook: Step-by-step procedures for reverting to non-AI systems during unexpected operational failure.

5. Tier-Dependent Approval Gate Authorities

The approval authority required to authorize an AI system directly corresponds to the risk tier established under AIG-002:

- LOW Risk Tier: Approved jointly by Technical Lead and Direct Engineering Manager within seven (7) business days.

- MEDIUM Risk Tier: Approved jointly by Department Vice President, Lead AI Engineer, and Information Security representative within fourteen (14) business days.

- HIGH Risk Tier: Requires unanimous approval by the Enterprise AI Ethics and Governance Board (AIEGB), CISO, and Chief Legal Officer following formal oral presentation and defense.

- CRITICAL Risk Tier: Requires unanimous ratification by the Board of Directors Risk Committee, CEO, CISO, and Chief AI Officer.

6. Specialized Approval Criteria for RAG and Autonomous Agents

RAG systems and agentic AI architectures introduce specialized failure modes requiring dedicated technical approval checks:

- Chunking and Embedding Validation: The evaluation team must verify that document chunking preserves semantic coherence, that embedding models do not introduce vector distortion, and that chunk headers contain mandatory metadata attributes (classification, department, allowed_roles under SEC-005).

- Retrieval Filtering Verification: Systems must be subjected to simulated permission escalation tests, validating that queries submitted by unprivileged user accounts never retrieve CONFIDENTIAL or RESTRICTED chunks.

- Grounding and Citation Verification: RAG applications must enforce answer grounding thresholds (minimum 90% citation alignment), ensuring the generator does not emit unsupported parametric assertions.

- Agentic Tool Sandboxing: Autonomous agents with API invocation capabilities must restrict execution privileges to read-only endpoints, with write actions requiring explicit human approval under AIG-005.

7. Technical Review Gateways: Benchmark Suites and Safety Thresholds

The Technical Review Board enforces strict quantitative thresholds during Stage 3 evaluation:

- Minimum Accuracy and F1 Score: Predictive models must exceed baseline domain benchmarks by at least 15% with a minimum F1 score of 0.85 on out-of-sample test splits.

- Latency Budgets: Real-time inference latency must remain under 350 milliseconds at the 95th percentile (P95) and under 600 milliseconds at the 99th percentile (P99) under simulated peak enterprise load.

- Adversarial Robustness Ratio: Generative models and prompt interfaces must withstand at least 95% of standardized automated jailbreak attempts and indirect injection payloads generated by enterprise red-team test suites.

- Hallucination Benchmark: Generative models must achieve a hallucination rate lower than 3.5% across domain-specific evaluation corpora, verified through automated semantic consistency audits.

- Disparate Impact Ratio: For any system influencing workforce hiring, compensation, or performance appraisal, the selection rate for any protected demographic group must not fall below 80% of the rate for the highest group (the 4/5ths rule).

8. Disposition Determinations: Approval, Conditional Approval, and Rejection

Upon reviewing the dossier and technical test results, the governing authority issues one of three formal determinations:

1. Unconditional Approval: The model satisfies all technical, ethical, security, and privacy criteria. Granted for a maximum duration of twelve (12) calendar months.

2. Conditional Approval: The model satisfies core safety baselines but exhibits minor non-critical deficiencies (e.g., minor documentation gaps, marginal latency overhead). The model may deploy to limited canary user groups under strict, time-bounded remediation conditions (maximum 60 days). Failure to satisfy remediation conditions results in automatic revocation.

3. Formal Rejection: The model exhibits unacceptable safety risks, unmitigated bias, severe prompt injection vulnerabilities, unverified data provenance, or negative ethical impact. Rejection is final for the submitted architecture.

9. Model Re-Approval Triggers and Governance Invalidation

An approved AI model does not retain permanent operational authorization. Re-approval through the full governance workflow is automatically triggered by any of the following events:

- Substantive Fine-Tuning: Modifying model weights through retraining or fine-tuning with newly acquired datasets.

- Knowledge Base Structural Updates: In RAG systems, introducing major new knowledge partitions or expanding access to higher classification data tiers.

- Performance or Data Drift: Detection of statistically significant input distribution drift or performance degradation under AIG-008.

- Security Incident Occurrence: The occurrence of a Severity 1 or Severity 2 AI Security Incident under AIG-009 automatically suspends model approval pending re-certification.

- Annual Expiration: All approvals expire automatically after 365 calendar days, requiring annual recertification audits.

10. Decommissioning, Graceful Degradation, and Model Retirement

When a model is superseded by a newer version or permanently retired, the project owner must execute the formal Decommissioning Runbook:

- User Notification: Downstream application teams and end users must receive thirty (30) days advance notice of model deprecation.

- Cache and Index Invalidation: Semantic search caches, vector embedding partitions, and model weights must be archived to cold storage and wiped from active execution clusters.

- Audit Registry Closure: The model status in the Enterprise AI Registry is formally updated to SUPERSEDED or ARCHIVED.

11. Audit Records, Governance Compliance, and References

All approval submissions, review notes, dissenting opinions, test reports, and formal sign-offs are archived in the Enterprise Governance Registry in tamper-evident storage for a minimum of seven (7) years.

This policy operates in direct alignment with related enterprise governance standards:

- AIG-002: AI Risk Classification Framework (Sets risk tiers governing approval gates).

- AIG-004: Generative AI Usage Policy (Defines approved enterprise use cases).

- AIG-006: AI Data Privacy Policy (Governs data handling reviews).

- AIG-007: Prompt Security Policy (Governs security validation testing).

- AIG-010: AI Model Deployment Policy (Translates approval into authorized production release).